

Benjamin Toll
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Professional Summary

Programmer, infrastructure specialist and strong debugger with 15+ years building reliable systems and testing infrastructure. Expert at identifying root causes in complex distributed systems and building deterministic CI/CD pipelines. Proven track record fixing broken test frameworks and enabling teams through automation and clear documentation.

Technical Skills

Testing & Debugging - Ginkgo, Go test framework, remote debugging (Delve), deterministic test design, identifying and fixing flaky tests, end-to-end testing

CI/CD & Automation - Jenkins (declarative pipelines, Groovy), GitHub Actions, GitLab, Buildkite, CircleCI, Prow, shell scripting, infrastructure automation

Languages - Go, Bash, Python, C, JavaScript, Ruby

Infrastructure & Tools - Linux, Kubernetes, OpenShift (ROSA), containers (Podman, Docker), Ansible, AWS

Professional Experience

Consultant
Software Engineer | March 2026 - Present

I've been contracted to improve test coverage, build pipelines and address technical debt. Mostly Golang and shell scripting.

Red Hat
Senior Software Engineer (Infrastructure) and Team Lead | October 2024 - September 2025

- Built reproducible remote debugging infrastructure (Delve, custom images, port-forwarding workflows) enabling the team to validate operators in live clusters without guesswork.
- Automated cluster provisioning with shell scripts implementing idempotency checks, state polling, dependency validation, and clear failure logging; reduced manual setup errors and enabled rapid, repeatable cluster creation and other operations.
- Eliminated race conditions and non-deterministic behavior in test suites and CI/CD builds and release pipelines; deep experience with end-to-end testing in complex multi-component environments.
- Comfortable with low-level troubleshooting (serial ports, noisy terminal output, hardware interaction) from prior embedded systems work.

Veriforce

Principal DevOps Engineer | December 2023 – October 2024

- Led GitOps transition for Kubernetes infrastructure; designed and documented deployment validation workflows that reduced rollback incidents by catching configuration errors before production.
- Evaluated Flux and Argo, built a business case for Flux based on our architecture and constraints. Wrote Go tooling using Kubernetes apimachinery libraries to detect configuration drift between live clusters (developers were making changes through Rancher UI) and state in GitHub, and generated YAML migrations—solving the critical problem of moving from manual configuration to source of truth without losing existing work.
- Designed and maintained CI/CD pipelines using Jenkins and GitHub Actions for microservices deployments on Kubernetes.
- Automated infrastructure provisioning and configuration management using Ansible and shell scripting.

Taos/IBM

Senior Software and DevOps Engineer | June 2021 – October 2022

- Built and maintained CI/CD pipelines using Jenkins and Kubernetes for distributed systems; designed validation workflows to catch configuration drift and deployment errors.
- Automated infrastructure provisioning and cluster management, reducing manual operational overhead.

Algorand

DevOps Engineer | November 2019 – February 2021

- Designed CI/CD pipelines using Jenkins and Kubernetes for distributed blockchain systems; built Go tooling to validate deterministic behavior across test environments.
- Conducted code reviews and enhanced security measures across application deployments.

Consultant

Software Engineer | January 2016 – October 2022

Full-stack web development and infrastructure consulting using modern web technologies and automation practices.

Sencha Inc

Software Engineer | June 2011 – December 2015

- Maintained and improved the Ext JavaScript framework, implementing new features and resolving critical bugs.
- Contributed to internal tooling and developer productivity enhancements.

Yahoo!

Software Engineer | November 2009 – May 2011

- Developed JavaScript applications for Yahoo's Connected TV platform.
- Served as principal maintainer for high-profile applications.

Connected TV platform development; hands-on experience with hardware interaction, firmware flashing, and embedded systems diagnostics (serial ports, low-level debugging).

Core Strengths

Root cause diagnosis in complex systems. Building repeatable testing frameworks. Clear technical communication. Debugging infrastructure at scale. Translating IT infrastructure practices to constrained environments.

Education

University of New Mexico
BA, History

Honors: Summa Cum Laude, Phi Beta Kappa, Phi Alpha Theta

Certifications

- Certified Kubernetes Administrator (CKA)
- Kubernetes and Cloud Native Associate – The Linux Foundation
- Linux Essentials (LE-1) – Linux Professional Institute

Languages

- English (Native/Fluent)
- Italian (Conversational)